

AI Research & Document Intelligence

Project Synopsis

1. Introduction

AI Research & Document Intelligence is an AI-powered research assistant that helps users interact with research papers and other PDF documents using natural-language questions. The system combines document processing, semantic search, vector databases, Retrieval-Augmented Generation (RAG), and a Large Language Model (LLM) to provide relevant answers from uploaded documents. It also displays source document and page information so that users can trace the information used for an answer.

2. Problem Statement

Research papers and technical documents often contain large amounts of information, making it difficult and time-consuming to manually locate specific information. Traditional keyword-based searching may not understand the meaning of a user's question. Therefore, there is a need for an intelligent system that can understand questions, retrieve semantically relevant sections from documents, and generate concise answers grounded in the provided research material.

3. Aim

To develop an AI-based document intelligence system that allows users to upload research PDFs, ask questions in natural language, retrieve relevant document content using semantic similarity, and generate clear answers with source and page references.

4. Objectives

1. Allow users to upload and process multiple PDF research documents.
2. Extract and clean text from uploaded PDF documents.
3. Divide document text into meaningful chunks for efficient retrieval.
4. Generate semantic embeddings using Sentence Transformers.
5. Store and search document embeddings using FAISS.
6. Retrieve the most relevant document chunks for a user question.
7. Use Gemini LLM with RAG to generate answers based on retrieved context.
8. Display document and page information for retrieved sources.
9. Provide recommended questions and maintain chat history.
10. Allow users to download a research report containing the question, answer, and sources.

5. Use Case

The system can be used by students, researchers, teachers, and professionals who need to quickly understand research papers or technical PDF documents. A user uploads one or more research papers, asks a question such as "What is the main objective of this research?", and receives an answer

generated from the relevant document content.

6. System Workflow

User → Upload Research PDF → PDF Text Extraction → Text Cleaning → Text Chunking → Embeddings → FAISS Vector Database → User Question → Question Embedding → Similarity Search → Relevant Chunks → Gemini LLM → Answer + Sources → User

7. Methodology

Component	Purpose
PDF Processing	Extracts text page-by-page from uploaded PDF documents.
Text Cleaning	Cleans extracted text before further processing.
Text Chunking	Divides documents into smaller chunks suitable for retrieval.
Embeddings	Converts text chunks and questions into numerical semantic vectors using Sentence Transformers.
Vector Search	FAISS stores the vectors and performs similarity search to identify relevant chunks.
RAG	Retrieved document chunks are supplied as context to the LLM.
Answer Generation	Gemini generates a concise answer using only the supplied research context.
Source Display	The application shows document name and page information for retrieved sources.

8. Technologies / Tech Stack

- Programming Language: Python
- User Interface: Streamlit
- PDF Processing: PyMuPDF (fitz)
- Text Processing: Python-based cleaning and chunking
- Embeddings: Sentence Transformers – all-MiniLM-L6-v2
- Vector Database/Search: FAISS
- Large Language Model: Gemini
- Architecture: Retrieval-Augmented Generation (RAG)
- Development Environment: VS Code

9. Expected Output

The application provides an interactive web interface where users can upload research papers, ask questions, select recommended questions, and receive AI-generated answers. Retrieved sources are displayed with document and page information. The user can also view previous questions and answers through chat history and download a research report containing the answer and source content.

10. Advantages

- Reduces the time required to search large research documents.
- Supports semantic rather than only keyword-based retrieval.
- Can process multiple PDF documents.
- Provides context-grounded answers through RAG.
- Shows source and page information for better traceability.
- Provides a simple and user-friendly Streamlit interface.
- Supports report export for later reference.

11. Limitations

The quality of answers depends on the quality and readability of the uploaded PDF documents, the retrieved chunks, and the language model. The current system is primarily designed for text-based PDF research documents; complex scanned documents, images, tables, and highly specialized layouts may require additional processing.

12. Future Scope

- Add OCR support for scanned research papers.
- Improve table, image, and figure understanding.
- Add stronger citation verification and citation links.
- Support more document formats such as DOCX and web pages.
- Add research-paper comparison and summarization features.
- Add user authentication and persistent document collections.
- Deploy the application as a cloud-based research assistant.
- Add multilingual question-answering support.

13. Conclusion

AI Research & Document Intelligence demonstrates how Generative AI, semantic embeddings, vector search, and RAG can be combined to build a practical document-question answering system. The project provides a useful solution for quickly extracting information from research documents while maintaining source awareness. It also demonstrates practical skills in Python, NLP, vector databases, LLM integration, and AI application development.

14. One-Line Synopsis

An AI-powered RAG-based research assistant that understands questions, retrieves relevant information from PDF documents using FAISS semantic search, and generates source-aware answers using Gemini.